

Employment

Product Engineer Intern	Isomer AI	May 2025 – Present
- Ship 10+ production TypeScript features end-to-end: real-time event pipelines (Pusher), Temporal.io distributed orchestration, Auth0 RBAC, multi-model AI integrations; 394 commits, 7 deployments. - Own backend stack (PostgreSQL, Redis, Docker, Terraform, Sentry); maintain CI/CD and resolve production incidents.		
Graduate Research Assistant	NC State University	April 2025 – Present
- Deploy genomic platform on GCP (Cloud Run) with JBrowse 2 genome browser; support 100+ researchers, 500GB+ datasets. - Improve ML model accuracy by 12% via hyperparameter tuning and time-series analysis on biological sensor data.		
Software Developer Intern	Mphasis (HP Company)	Jun 2024 – Aug 2024
- Built 5+ AI workflows including health claim automation (OCR, NLP, ML); improved document processing accuracy by 25%. - Automated AWS and GCP deployments via CI/CD; integrated Grafana monitoring and led model red-teaming for reliability.		
AI/ML Intern	Capgemini	Jun 2023 – Aug 2023
- Led SDLC for data-driven ticket analysis system using NLP classification; reduced average resolution time by 15%. - Built Kafka + Spark + Hadoop pipelines for batch and real-time data ingestion; cut ticket-processing latency by 20%.		

Education

Raleigh, NC	North Carolina State University	Aug 2024 – May 2026
- Master of Science in Computer Science - Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering.		
Mumbai, India	Mumbai University	Aug 2020 – May 2024
Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infra, DS & Algorithms, Machine Learning, Blockchain.		

Projects

KubeServe	Kubernetes LLM Inference Platform	2026
- Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing. - Instrumented OTel + Prometheus + Grafana observability; Helm + ArgoCD GitOps; k6 to 500 VUs; 88 pytest tests.		
AlgoPulse	Agentic AI Platform	2025
- Built agentic AI orchestration platform on Temporal.io and Kafka supporting multi-step reasoning pipelines; sustained 2,000 async inference requests per second under sustained production load. - Replaced API polling with Supabase Edge Functions for real-time AI streaming; eliminated state loss under peak traffic.		
PulseRank	Real-Time Leaderboard & Activity Feed	2026
- Built event-driven leaderboard and activity feed in TypeScript with NestJS: MySQL transactional writes, MongoDB TTL projections, Redis sorted sets for sub-millisecond ranked reads; SSE live updates. - Validated with Testcontainers; benchmarked hot paths with k6; OpenTelemetry + Prometheus; GitHub Actions CI.		
Demand Paging	Xinu OS – Systems Programming	2025
- Implemented 8-store virtual memory system for Xinu OS in C: demand paging, page table management, Second-Chance page-replacement algorithm, address-space TLB fault handler; cut page fault rate by 25%. - Engineered 7+ kernel syscalls (address translation, TLB, frame allocation); improved memory throughput by 15%.		

Languages and Technologies

- Languages:** C, Python, Java, TypeScript, JavaScript, SQL, Bash
- Systems & Infra:** Linux, Kubernetes, KEDA, Docker, Kafka, Temporal.io, Redis, OpenTelemetry, Prometheus
- Backend & Data:** PostgreSQL, MySQL, MongoDB, Node.js, NestJS, FastAPI, REST APIs, k6, Testcontainers
- Cloud & DevOps:** GCP, AWS, Cloud Run, Terraform, Helm, ArgoCD, GitHub Actions, Grafana, Sentry

Additional Experience and Awards

- IEEE Published:** Co-authored paper on CV-Based Cybersecurity Threat Detection (Springer icSoftComp 2023); presented to 500+ attendees.
- National Hackathon Winner:** Won or ranked Top 10 in 6+ national hackathons spanning ML, systems, and developer tools; built 3 complete MVPs end-to-end within 48-hour sprints, each with a working demo.
- Lead at Google Developers Student Club:** Led 4+ tech events (hackathons, conferences) reaching 500+ students.
- Technical Writing:** Published on LLM model collapse, AI agent security, audio AI at kulkarniaditya.com/blog